

SAGRES, Portugal

WMO: 085330

Lat: 37.0128N

Lon: 8.9489W

Elev: 26

StdP: 101.02

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	5.9	-2.3	3.1	7.8	0.4	3.9	8.5	12.4	12.6	11.0	12.8	2.0	70	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	28.9	19.7	27.2	19.5	25.7	19.2	21.4	25.3	20.7	24.5	20.2	23.7	3.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	15.0	22.7	19.6	14.4	22.1	19.1	13.9	21.7	62.3	25.3	59.9	24.7	58.1	23.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.9	9.1	2.1	32.6	1.7	2.7	0.9	34.6	-0.1	36.2	-1.0	37.7	-2.3	39.7		
			0.5	22.6	1.7	0.9	-0.8	23.2	-1.8	23.7	-2.8	24.2	-4.0	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	12.2	12.6	14.3	15.5	17.6	19.8	20.7	21.3	20.5	18.6	15.2	13.3
	DBStd	3.79	2.33	2.13	1.91	1.75	2.05	1.99	1.77	1.89	1.60	1.97	2.40	2.37
	HDD10.0	16	7	4	1	0	0	0	0	0	0	0	1	2
	HDD18.3	884	189	161	126	87	38	6	1	0	2	20	97	157
	CDD10.0	2504	76	77	133	164	236	294	331	350	315	267	156	104
	CDD18.3	332	0	0	0	1	16	51	73	92	67	29	3	0
	CDH23.3	1070	0	0	1	6	51	165	252	362	186	47	0	0
CDH26.7	206	0	0	0	0	8	29	57	87	22	3	0	0	
Wind	WSAvg	4.3	3.7	4.0	4.0	4.3	4.6	5.0	5.9	5.3	3.8	3.4	3.6	3.8
Precipitation	PrecAvg	461	80	68	44	37	19	8	0	4	13	55	65	66
	PrecMax	708	169	208	105	97	66	50	1	47	71	213	213	208
	PrecMin	255	0	1	2	2	0	0	0	0	0	6	1	1
	PrecStd	121	52	47	28	30	17	12	0	11	21	62	54	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	18.4	21.6	24.1	27.2	29.9	31.2	31.4	29.0	26.7	21.9	18.9
		MCWB	15.2	14.7	15.0	16.6	17.4	19.0	19.5	19.8	19.0	18.4	18.1	16.3
	2%	DB	17.3	17.4	19.6	21.6	24.4	26.9	28.1	28.8	26.9	24.4	20.9	18.2
		MCWB	14.9	14.4	14.7	15.6	17.2	18.9	19.8	20.1	19.6	18.8	17.7	16.3
	5%	DB	16.6	16.7	18.5	20.1	22.8	25.0	26.1	27.1	25.4	23.0	20.0	17.7
		MCWB	14.4	14.0	14.6	15.5	16.8	18.7	19.3	20.2	19.3	18.9	17.2	15.9
	10%	DB	16.1	16.1	17.6	19.0	21.2	23.5	24.2	25.1	24.0	22.0	19.2	17.1
		MCWB	13.9	13.7	14.4	15.3	16.7	18.4	19.0	19.6	19.2	18.4	16.6	15.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	16.1	16.8	18.1	19.7	21.2	21.6	22.7	22.1	21.3	19.9	17.7
		MCDB	17.7	17.2	18.8	21.3	23.3	24.8	26.9	26.9	25.6	23.0	21.1	18.2
	2%	WB	15.8	15.4	16.1	17.0	18.6	20.2	20.6	21.5	21.2	20.4	18.9	17.1
		MCDB	16.6	16.6	17.9	19.8	22.0	24.4	25.6	25.9	24.3	22.3	20.3	17.9
	5%	WB	15.1	14.8	15.6	16.2	17.9	19.5	20.0	20.7	20.5	19.7	18.0	16.4
		MCDB	16.2	16.2	17.4	18.9	21.0	23.5	24.1	25.1	23.5	21.8	19.4	17.5
	10%	WB	14.4	14.2	15.1	15.6	17.2	18.9	19.4	20.1	19.9	19.1	17.1	15.6
		MCDB	15.8	15.8	16.9	18.3	20.2	22.5	23.2	24.0	22.8	21.2	18.7	16.9
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.4	6.3	6.1	6.3	6.7	6.8	7.1	6.9	6.3	6.6	6.3
		MCDBR	6.6	6.6	7.6	8.0	9.0	9.4	9.9	10.2	9.3	7.8	6.4	5.6
		MCWBR	5.0	4.7	4.8	4.8	4.5	4.0	4.3	4.7	4.7	4.6	4.5	4.4
	5% WB	MCDBR	4.4	5.3	5.9	6.7	6.8	7.6	7.9	8.3	7.3	6.1	4.7	3.9
		MCWBR	4.0	4.2	4.3	4.4	3.9	3.8	3.8	4.3	4.4	4.1	3.9	3.7
Clear-Sky Solar Irradiance	taub	0.336	0.351	0.385	0.379	0.386	0.390	0.377	0.396	0.389	0.375	0.349	0.340	
	taud	2.514	2.475	2.380	2.421	2.426	2.430	2.453	2.419	2.454	2.492	2.523	2.505	
	Ebn at Noon	852	882	878	903	897	889	898	876	864	844	830	820	
	Edh at Noon	80	94	113	114	115	115	111	113	104	91	79	77	
All-Sky Solar Radiation	RadAvg	2.56	3.54	4.78	6.00	7.06	7.56	7.67	6.91	5.56	3.96	2.81	2.30	
	RadStd	0.27	0.38	0.39	0.34	0.38	0.40	0.26	0.23	0.29	0.28	0.27	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page